

## **Контрольная работа №2**

### **Вариант 5**

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# Контрольная работа 2<sup>ч</sup> Вариант 5

$$1) (A \wedge \bar{B}) \vee (A \wedge B) \vee (A \wedge \bar{C}) = \\ = A(\bar{B} \vee B \vee \bar{C}) = A(1 \vee \bar{C}) = A$$

$$2) (\overline{\bar{X} \wedge \bar{Y} \vee \bar{X}}) \wedge (\overline{X \vee \bar{X} \wedge Y}) = \\ = (\bar{X} \vee \bar{Y} \vee \bar{X}) \wedge (\overline{X \vee \bar{X} \vee \bar{Y}}) = \\ = (1 \vee \bar{Y}) \wedge (\bar{X} \wedge \bar{X} \wedge Y) = \bar{X} \wedge Y$$

$$3) (A \vee C) \wedge (\bar{A} \vee B) \wedge (A \vee \bar{C}) = \\ = (AA \vee A\bar{C} \vee AC \vee C\bar{C}) \wedge (\bar{A} \vee B) = \\ = (A \vee A\bar{C} \vee AC) \wedge (\bar{A} \vee B) = \\ = A(1 \vee \bar{C} \vee C) \wedge (\bar{A} \vee B) = A \wedge (\bar{A} \vee B) = \\ = A\bar{A} \vee AB = AB$$

$$4) (A \oplus C) \wedge (\bar{A} \vee B) \wedge (A \vee \bar{C}) = \\ = (A\bar{C} \vee C\bar{A}) \wedge (\bar{A} \vee B) \wedge (A \vee \bar{C}) = \\ = (A\bar{C} \wedge (\bar{A} \vee B) \vee C\bar{A} \wedge (\bar{A} \vee B)) \wedge (A \vee \bar{C}) = \\ = (0 \vee A\bar{C}B \vee C\bar{A} \vee C\bar{A}B) \wedge (A \vee \bar{C}) = \\ = (A\bar{C}B \vee C\bar{A}) \wedge (A \vee \bar{C}) = A\bar{C}B \vee 0 \vee A\bar{C}B \vee 0 =$$

$$= A \bar{C} B$$

$$5) A \vee (X \downarrow C) \vee \bar{C} = A \vee (\bar{X} \wedge \bar{C}) \vee \bar{C} =$$

$$= A \vee \bar{C} \wedge (1 \vee \bar{X}) = A \vee \bar{C}$$

$$F(A, B, C) = (B \wedge A) \vee (\overline{B \wedge C}) =$$

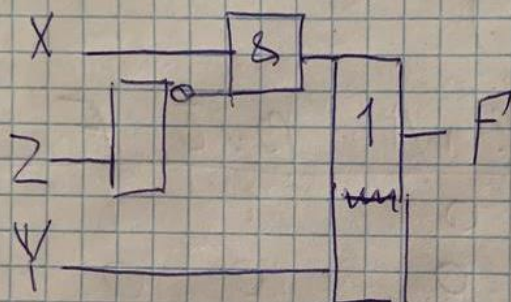
$$= (B \wedge A) \vee (B \vee \bar{C})$$

| A | B | C | $B \wedge A$ | $B \vee \bar{C}$ | F |
|---|---|---|--------------|------------------|---|
| 0 | 0 | 0 | 0            | 1                | 1 |
| 0 | 0 | 1 | 0            | 0                | 0 |
| 0 | 1 | 0 | 0            | 1                | 1 |
| 0 | 1 | 1 | 0            | 1                | 1 |
| 1 | 0 | 0 | 0            | 1                | 1 |
| 1 | 0 | 1 | 0            | 0                | 0 |
| 1 | 1 | 0 | 1            | 1                | 1 |
| 1 | 1 | 1 | 1            | 1                | 1 |

) =

3

$$\begin{aligned} \text{C.D.H.P.} &= \bar{X}\bar{Y}\bar{Z} + \bar{X}Y\bar{Z} + \bar{X}YZ + \\ &+ X\bar{Y}\bar{Z} + XY\bar{Z} = \bar{X}(\bar{Y}\bar{Z} + Y\bar{Z} + YZ) + \\ &+ X(\bar{Y}\bar{Z} + Y\bar{Z}) = \bar{X}(\bar{Y}\bar{Z} + Y) + \bar{X}(Y\bar{Z} + YZ) \\ &= Y + X\bar{Z} \end{aligned}$$



4

$$V_1 = 1010101010100010$$

$$V_2 = 0000000000$$

$$w(V_1) = 8 \quad w_2(V_2) = 0$$

5

$$V_1 = 1010101010100000$$

$$V_2 = 0000100100110010$$

$$d(V_1, V_2) = 7$$

$\overbrace{1 \dots 6}^1$

(2)

$$d \leq t \quad (t=1)$$

$$d(V_1, V_0) = 1$$

$$d(V_2, V_0) = 1$$

$$d(V_3, V_0) = 2$$

$$d(V_4, V_0) = 1$$

$$d(V_5, V_0) = 2$$

$$d(V_6, V_0) = 2$$

$$d(V_7, V_0) = 3$$

$$G = \{V_0\}$$

$$G = \{V_0, V_3\}$$

$$d(V_5, V_3) = 2$$

$$d(V_6, V_3) = 2$$

$$d(V_7, V_3) = 1$$

$$d(V_8, V_5) = 2$$

$$G = \{V_0, V_3, V_5\} \quad G = \{V_0, V_3, V_5, V_6\}$$

Omben: 110000101011

$\overbrace{1 \dots 7}^1$

$$d \leq t \quad (t=2)$$

$$d(V_1, V_0) = 1$$

$$d(V_2, V_0) = 1$$

$$d(V_3, V_0) = 2$$

$$d(V_4, V_0) = 1$$

$$d(V_5, V_0) = 2$$

$$d(V_6, V_0) = 2$$

$$d(V_7, V_0) = 3$$

$$d(V_8, V_0) = 1$$

$$d(V_9, V_0) = 2$$

$$d(V_{10}, V_0) = 2$$

$$d(V_{11}, V_0) = 3$$

$$d(V_{12}, V_0) = 2$$

$$d(V_{13}, V_0) = 3$$

$$d(V_{14}, V_0) = 3$$

$$d(V_{15}, V_0) = 4$$

$$G = \{V_0\}$$

$$G = \{V_0, V_{15}\}$$

$$d(V_{11}, V_{15}) = 1$$

$$d(V_{13}, V_{15}) = 1$$

$$d(V_{14}, V_{15}) = 1$$

$$d(V_7, V_{15}) = 1$$

$$G = \{V_0, V_{15}\}$$

Пример: 01110000

Когда не совсем выполняется условие м.к.  $d(V_0, V_{15}) = 4 < 2t + 1 = 5$

8

$$2^k \geq M + k + 1$$

$$M = 7$$

$$2^k \geq 8 + k$$

$$2^1 \geq 8 + 1 \text{ не}$$

$$2^2 \geq 8 + 2 \text{ не}$$

$$2^3 \geq 8 + 3 \text{ не}$$

$$2^4 \geq 8 + 4 \text{ да}$$

$$\underline{k=4}$$

~~8~~

$$M = 23$$

$$2^k \geq 24 + k$$

$$2^5 \geq 24 + 5$$

$$\underline{k=5}$$

$$M = 44$$

$$2^k \geq 44 + k$$

$$2^6 \geq 44 + 6 \text{ да}$$

$$\underline{k=6}$$

$\nearrow 9$

$$M = 10001$$

$$N = M + k = 9$$

$$2^k \geq 6 + k$$

$$k = 4$$

| $M$   |  | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Row-60 | But |
|-------|--|---|---|---|---|---|---|---|---|---|--------|-----|
|       |  | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1      |     |
| $k_1$ |  |   |   | 1 |   |   |   |   |   | 1 | 2      | 0   |
| $k_2$ |  |   |   | 1 |   |   |   |   |   |   | 1      | 1   |
| $k_3$ |  |   |   |   |   |   |   |   |   |   | 0      | 0   |
| $k_4$ |  |   |   |   |   |   |   |   |   | 1 | 1      | 1   |

$$M \rightarrow 110000110$$

$$\begin{array}{r} 3 \quad 0 \quad 0 \quad 1 \quad 1 \\ 9 \quad 1 \quad 0 \quad 0 \quad 1 \\ \hline 1 \quad 0 \quad 1 \quad 0 \\ \hline k_4 \quad k_3 \quad k_2 \quad k_1 \end{array} \text{ xor}$$

$\nearrow 10$

| No             | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 |
|----------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|                | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1  | 0  | 0  | 0  | 1  | 1  | 0  | 1  | 0  | 0  | 1  | 1  | 0  | 1  | 1  | 1  | 0  | 1  | 1  | 1  |
| k <sub>1</sub> |   |   | 1 |   |   |   |   | 1 |   |    |    |    |    |    | 1  |    | 1  |    |    |    |    | 1  |    |    |    |    | 1  |    | 1  |
| k <sub>2</sub> |   |   | 1 |   |   |   | 1 | 1 |   |    | 1  |    |    | 1  | 1  |    |    |    |    |    | 1  | 1  |    | 1  |    |    |    | 1  |    |
| k <sub>3</sub> |   |   |   |   |   | 1 | 1 |   |   |    |    |    |    | 1  | 1  |    |    |    |    |    | 1  | 1  |    |    | 1  | 1  |    | 1  |    |
| k <sub>4</sub> |   |   |   |   |   |   |   |   |   | 1  |    |    |    | 1  | 1  |    |    |    |    |    |    |    |    |    | 1  | 1  |    | 1  |    |
| k <sub>5</sub> |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    | 1  |    |    | 1  | 1  |    | 1  | 1  | 1  |    | 1  |    | 1  |

$k_1 k_2 k_3 k_4 k_5$   
 $10100 - \text{new}$   
 $01010 - \text{old}$   
 $\hline 11110_2 = 30_{10} \Rightarrow \text{number} > 1$

$\nearrow 11$

$$G = \left| \begin{array}{cc|cc} 100 & 110 & 1 & 00 \\ 010 & 011 & 0 & 10 \\ 001 & 101 & 0 & 01 \end{array} \right|$$

$100_2 = 4_{10} \Rightarrow 100110 - \text{neg для 4 в ППК}$   
 $110_2 = 6_{10} \Rightarrow 100110 \text{ xor } 010011 = 110101 -$   
 $\text{neg для 6 в ППК}$

$110101 \xrightarrow{\text{вводим единицу}} 111101$

| бит | Проверочная матрица |
|-----|---------------------|
| 0   | 1 0 1 1 0 0         |
| 0   | 1 1 0 0 1 0         |
| 1   | 0 1 1 0 0 1         |

$$S_1 = n_1 + n_3 + n_6 = 1$$

$$S_2 = n_1 + n_2 + n_5 = 0$$

$$S_3 = n_2 + n_3 + n_6 = 1$$

$S = 101$  - соответствует 3 столбцу  
в проверочной матрице или 3  
строке в матрице ошибок  $\Rightarrow$   
 $\Rightarrow$  ошибка в 3 позиции

$$M' = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \rightarrow \text{матрица ошибок}$$